

## Project Initialization and Planning Phase

|               |                                                         |
|---------------|---------------------------------------------------------|
| Date          | 29 September 2026                                       |
| Team ID       | UNM123APEXLAB                                           |
| Project Title | ComicCraft - AI Comic Story Creator using Gemini Models |
| Maximum Marks | 5 Marks                                                 |

### Project Proposal (Proposed Solution) report

The proposal report aims to make comic creation simple for everyone using generative AI. ComicCraft is a web-based application that turns a user-provided prompt into a complete comic story with illustrations, using Google Gemini models. Key features include AI story generation, AI illustration generation and a simple web interface.

| Project Overview  |                                                                                                                                                                                                                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | The primary objective is to let anyone create personalized comic book stories and illustrations from a simple text prompt, using Gemini AI models, without needing drawing or design skills.                                                                                                |
| Scope             | The project covers a web application built with FastAPI that accepts user prompts, generates a panel-wise story and matching illustrations through Gemini models, and displays the finished comic to the user.                                                                              |
| Problem Statement |                                                                                                                                                                                                                                                                                             |
| Description       | Creating a comic needs both storytelling and illustration skills, and the process is slow and costly. Most people with good story ideas cannot turn them into visual comics.                                                                                                                |
| Impact            | Solving this saves time and cost, makes comic creation accessible to students, writers and creators, and encourages creativity and visual storytelling.                                                                                                                                     |
| Proposed Solution |                                                                                                                                                                                                                                                                                             |
| Approach          | Using the Google Gemini text model to write the story and scene descriptions, and Gemini image generation to illustrate each panel, served through a FastAPI backend and a web front end.                                                                                                   |
| Key Features      | <ul style="list-style-type: none"><li>- Prompt-based comic story generation using Gemini models.</li><li>- AI-generated illustrations for every comic panel.</li><li>- Easy-to-use web interface built on FastAPI.</li><li>- Personalized stories by genre, style and characters.</li></ul> |

### Resource Requirements

| Resource Type       | Description                             | Specification/Allocation                                |
|---------------------|-----------------------------------------|---------------------------------------------------------|
| Hardware            |                                         |                                                         |
| Computing Resources | CPU/GPU specifications, number of cores | Standard CPU (no local GPU needed; Gemini runs via API) |
| Memory              | RAM specifications                      | 8 GB                                                    |

|                         |                                       |                                                                                        |
|-------------------------|---------------------------------------|----------------------------------------------------------------------------------------|
| Storage                 | Disk space for data, models, and logs | 256 GB SSD                                                                             |
| <b>Software</b>         |                                       |                                                                                        |
| Frameworks              | Python frameworks                     | FastAPI, Uvicorn                                                                       |
| Libraries               | Additional libraries                  | google-generativeai (Gemini SDK), Pydantic, Pillow, Jinja2                             |
| Development Environment | IDE                                   | VS Code, Jupyter Notebook                                                              |
| <b>Data</b>             |                                       |                                                                                        |
| Data                    | Source, size, format                  | No fixed dataset; user text prompts as input and Gemini API for story and image output |